a

Human decides

- **Scientist** — asks the question and scopes it in time
- **Research contract** — focus terms, forbidden equivalences, tracks, required gates, acceptance criteria and stop rule, hashed before the first task
- A human writes no claim or hypothesis; an agent grades nothing
- **Expert** — alone sets G7, closes interpretive objections, approves changes to canonical knowledge

Agents propose

Plan and reflect

- **Director** C K A X ● — plans rounds, picks agents
- **Meta-reviewer** C K ● — weak points, next round

a new round until the contract's stop rule is met

Read (from round 0)

- **Curator** C ○ — freezes scope and coverage
- **Lineage** C A ○ — citation and transcription links
- **Philologist** C ● — readings, variants, collation
- **Scholar** C ○ — textual history of focus terms
- **Semanticist** C K ● — period-bound senses of terms
- **Evidence** C K ● — verbatim, locatable evidence
- **Extractor** C K ● — claims with verbatim spans

Discover (from round 1)

- **Pattern miner** C K A ○ — discovery engines D1–D5
- **Statistician** A ○ — tests, FDR, coverage
- **Evolution** A ○ — concept and formula timelines
- **Hypothesis** C K X ● — testable hypotheses, revisions

Criticize

- **Skeptic** C K A X ● — attacks every hypothesis

objections go back to the Hypothesis agent, which may only narrow the claim

Translate (if asked)

- **Ontologist** C K P ● — ancient–modern mappings
- **Modern evidence** C K L ● — sole door to modern data

historical hypotheses mark G8 not applicable

▼ BeforeClaimCommit: provenance · semantic support · anachronism ▼ AfterHypothesis: anachronism ▼ BeforeGraphWrite: anachronism

Kernel records

Append-only event log (61 event types)

every event carries its principal, and the reducer refuses one that its principal may not emit (b); state is replayed and hashed every round; budgets, checkpoints and the stop rule are the kernel's

▼ BeforeArtifactPublish: publish guard

Validators grade

- **Gates G0–G8** — per claim, evidence and hypothesis; G7 only by the expert
- **Claim validation** — the passage exists, the quote is verbatim at its span, and the quote states the claim; otherwise the claim is rejected
- **Scores and ranks** — discovery score and Elo; a ranking, never a truth value
- **Objections** — a hard one closes only when its check re-runs and passes
- **Contract verdict** — accepted, needs expert, inconclusive or rejected

Tools and data

- **Corpus store** — dated, layered witnesses, read only through scoped views (past views for blind tests)

Tool mesh: 73 tools, each behind one permission; every call passes the BeforeToolCall scope guard and the AfterToolCall result firewall

retrieval 5 C — search, passages, lineage	**study 35** C — the study layer	**philology 6** C — collation, page images	**knowledge 4** K — claims, terms	**knowledge 1** P — propose a change	**analytics 9** A — D1–D5 engines
validation 4 C — quote checks, audit	**validation 1** V — gates	**bridge 3** C — ancient phenotypes	**literature 4** L — modern evidence	**research 1** X — sandboxed code	

● may consult a model where the rules leave a case uncertain ○ rules only Permissions: C classics:read · K kg:read · P kg:propose · A analysis:run · L literature:read · X codemode:exec · V validation:run

Gates: G0 source, G1 transcription, G2 philology, G3 sense, G4 claim provenance, G5 independent sources, G6 falsification, G7 expert, G8 modern evidence

b

Events a principal may emit (count)	Agent	Validator	Human	Kernel
Plan and run tasks; scope the corpus; propose evidence, claims, lineage, observations; generate, review and revise hypotheses (21)	●	●	·	●
Call tools; record a decision; propose a change (7)	●	●	●	●
Call a model (1)	●	·	·	●
Grade: gates, scores, ranks, confirmation; evaluate the contract; publish (6)	·	●	·	●
Reject a hypothesis; retract evidence; write memory (3)	·	●	●	●
Close an objection; approve or reject a change (3)	·	●	●	·
Approve or reject a hypothesis (G7); steer a research (3)	·	·	●	·
Stop a research (1)	·	·	●	●
Open the contract, run rounds, spawn agents, route models, budget, checkpoint, complete (16)	·	·	·	●

Gate results: gates are graded by validators, never by the work they grade. Objections: an objection is closed by re-running its check (validator) or by an expert, never by a model. Only a human expert sets G7.

c

Jingui yaolüe ch. 13 on xiaoke 消渴 in three witnesses (punctuation removed)

omission (脱) 丸 → 圓

Deng Zhen print, 1340 男子消渴小便反多以飲一斗小便一斗腎氣丸主之
Wu Qian manuscript, 1395 男子消渴小便反多以飲一斗－－－－腎氣丸主之
Maijing, Yuan print, 1330 男子消渴小便反多以飲一斗小便一斗腎氣圓主之

圓 for 丸 avoids the name of Emperor Qinzong (桓, from 1126): that witness descends from an edition made after 1126

- **Claim** — Shenqi wan 腎气丸 indicated for 消渴 with polyuria
- **Quote** — 男子消渴，小便反多，以飲一斗，小便一斗，腎氣丸主之
- **Passage** — cm_jingui_dengzhen.00712
- **Locator** — 卷中 · 消渴小便利淋病脉證并治第十三 · p. 100
- **Witness** — Deng Zhen print, 1340; Peking University Library
- **Work · layer** — Jingui yaolüe, 200–219 · main text
- **Source · licence** — CMETA, collated page by page · CC BY 4.0
- **Check in chat** — ✓ verbatim ≠ quote differs × no such passage